

# IND-PV-SOP-001 — Sales Metric Definitions & KPI Dictionary

**Document ID:** IND-PV-SOP-001 **Version:** 2.1 **Effective date:** 01-Jan-2024 **Last review:** 15-Jun-2026 **Owner:** Sales Analytics COE / India Passenger Vehicles **Audience:** Executives, Business Controllers, Regional Managers, Data Analysts **Applies to dataset:** fact\_sales, dim\_carline, dim\_color, dim\_salesman, dim\_region (India PV 2019–2026)

## 1. Purpose

This SOP defines the **certified business metrics** used for India passenger-vehicle (PV) retail analytics. All dashboards, chat answers, executive packs, and narrative commentary must use these definitions so numbers remain comparable across regions, OEMs, and time periods.

## 2. Scope

- Retail sales orders for passenger vehicles in India (hatchback, sedan, SUV, MPV).
- Powertrains: Petrol, Diesel, Hybrid, Electric.
- Geographic coverage: North, West, South, East, Central India metros and hubs.
- Time window of the reference warehouse: **January 2019 through June 2026**.

Out of scope: commercial vehicles, two-wheelers, spare-parts aftermarket, and wholesale dealer stock (unless separately certified later).

## 3. Certified metrics (map to physical columns)

Business term — Certified definition — SQL / column rule — Format — Do NOT use

|---|---|---|---|---|

**Revenue** — Gross retail invoice value of vehicles sold — ``SUM(total_sales)`` — INR (₹ / Lakh / Crore) — ``price_per_unit`, `AVG(total_sales)``

**Units Sold** — Physical vehicle volume — ``SUM(order_qty)`` — Integer + “units” — ``COUNT(*)`` alone when volume is asked

**Orders** — Distinct sales transactions — ``COUNT(order_id)`` — Integer — Confusing with Units Sold

**Average Selling Price (ASP)** — Revenue per unit — ``SUM(total_sales) / NULLIF(SUM(order_qty),0)`` — INR — Average of ``price_per_unit`` without qty weight

**Average Order Value (AOV)** — Revenue per order — ``SUM(total_sales) / NULLIF(COUNT(order_id),0)`` — INR — Same as ASP

**Active Salespeople** — Distinct sellers with at least one order in the filtered period — ``COUNT(DISTINCT sales_person_id)`` — Integer — Headcount from HR master alone

**Revenue per Salesperson** — Productivity — ``SUM(total_sales) / NULLIF(COUNT(DISTINCT sales_person_id),0)`` — INR —

### 3.1 Ambiguity rule (“sales”)

If a user says “sales” without clarifying:

- Default interpretation = **Revenue** (``SUM(total_sales)``).
- If the question contains volume words (units, volume, cars sold, how many vehicles) → **Units Sold**.
- If the question contains “orders / bookings / transactions” → **Orders**.
- When still ambiguous, ask: Revenue / Units / Orders.

### 3.2 Top Model / Top Performer rule

For cards or questions labelled **Top Model**, **Top Performer**, or **Sales Volume**:

- Rank by ``SUM(order_qty)`` (units), never by revenue alone.

- Display as: `{model} — {N} units sold`.
- A low-priced high-volume hatchback must outrank a high-priced low-volume SUV on volume KPIs.

## 4. Time intelligence standards

Grain — Label format — Rule

|---|---|---|

Day — `YYYY-MM-DD` — `sales\_date`

Month — `YYYY-MM` — `strftime("%Y-%m", sales\_date)`

Quarter — `Q1-2023`, `Q2-2023`, ... — Month bands 1–3 / 4–6 / 7–9 / 10–12. **Never** float division `(month-1)/3`

Year — `YYYY` — Calendar year on `sales\_date`

### 4.1 Festive seasonality (India)

September–December typically carry festive uplift (Navratri, Diwali, year-end). Narratives must not treat Q4 strength as “unexpected” without checking festive baseline.

### 4.2 COVID baseline year

**2020 is a depressed baseline**, not a normal comparator. Prefer YoY vs 2019 or vs 2021+ when commenting on recovery. See IND-PV-SOP-002.

## 5. Dimension standards

Business concept — Preferred attribute — Notes

|---|---|---|

Brand / OEM — `make` — Maruti, Tata, Mahindra typically lead volume in this market

Model — `model` or `carline\_name` — Prefer friendly labels over IDs

Body style — `car\_type` — SUV / Hatchback / Sedan / MPV

Powertrain — `engine\_type` — Petrol / Diesel / Hybrid / Electric

Colour — `colour\_name` — White / Silver / Black dominate India preference

Region — `region\_name` + `city` — Never show bare `region\_id` to executives

Salesperson — `first\_name` — — — — `last\_name` — Never raw `sales\_person\_id` in narratives

## 6. Narrative quality bar (executives)

Every executive-facing narrative should include:

- **What happened** (metric + period + grain).
- **Who led / lagged** (make, region, or model — max 2 leaders).
- **Business context** (COVID, festive, EV shift — only if relevant).
- **Recommended next question** (drill path).

Avoid: dumping raw IDs, decimal quarter labels, “breakdown of revenue” without a leader, or currency in £ when domain is India (use ₹).

## 7. Data quality expectations

- `sales\_date` must be non-null after load (ISO dates). Null dates invalidate time narratives.
- `order\_qty` ≥ 1 on fact rows.

- EV rows store **battery kWh** in ``engine_capacity`` (not zero). Do not flag EV capacity as financial outliers.
- Join path: ``fact_sales`` LEFT JOIN all dimensions on foreign keys defined in the semantic model.

## 8. Change control

Metric definition changes require COE approval and glossary YAML update in the same release. Chat answers must follow glossary SQL expressions when present.